



COMSATS University Islamabad, Lahore Campus

Department of Computer Engineering

Quiz # 3

Course Title:	VLSI Design	Course Code:	CPE434	Credit Hours:	4(3,1)
Course Instructor:	Dr. Muhammad Naeem Awais	Programme Name:	BS Computer Engineering		
Semester:		Batch:		Date:	11 st May 2026
Time Allowed:	10 Minutes		Maximum Marks:	5	
Student's Name			Registration. No.:		

Question # 1:

(PLO3-CLO3-C5)

Design a circuit for the following function at transistor level with Dual-Rail Domino:

$$F(A,B,C,D) = \sum(0,2,4,5,9,11,12,15)$$

$$= \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}\bar{D} + \bar{A}B\bar{C}D + \bar{A}B\bar{C}D + A\bar{B}\bar{C}\bar{D} + A\bar{B}C\bar{D} + ABCD$$

